TAN TAO UNIVERSITY
FACULTY OF INFORMATION TECHNOLOGY

Report of Final Project
DECODING THE GLOBAL PLASTIC WASTE CRISIS
Course: Data visualization
Spring 2026

Group

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1. PROBLEM DEFINITION

In this project, we do not view plastic waste merely as an environmental issue but approach it through the lens of **"Management System Failure."**

- **Core Issue:** There is a critical disconnect between the exponential growth of the global plastic production industry and the developmental pace of waste management infrastructure.
- **Research Hypothesis:** Current marine plastic pollution is not solely caused by over-consumption, but is a consequence of plastic being produced in volumes that far exceed the absorption and recycling capacities of the current economy.
- **Objective:** Utilize data to identify "bottlenecks" in the plastic life cycle, thereby proposing priority areas for infrastructural intervention rather than focusing solely on individual behavioral changes.

2. DATASET SELECTION, PREPARATION, AND VISUALIZATION

2.1. Dataset

The project integrates data from four strategic sources from *Our World in Data* to ensure multidimensionality:

1. Production: Historical data from 1950 to 2019.
2. Plastic Fate: Breakdown by rates of recycling, incineration, landfilling, and mismanagement.
3. Domestic Management: Per capita mismanaged plastic waste (kg/year).
4. Oceanic Impact: Actual share of global plastic waste emitted to the ocean by country.

2.2. Data Preparation & Cleaning

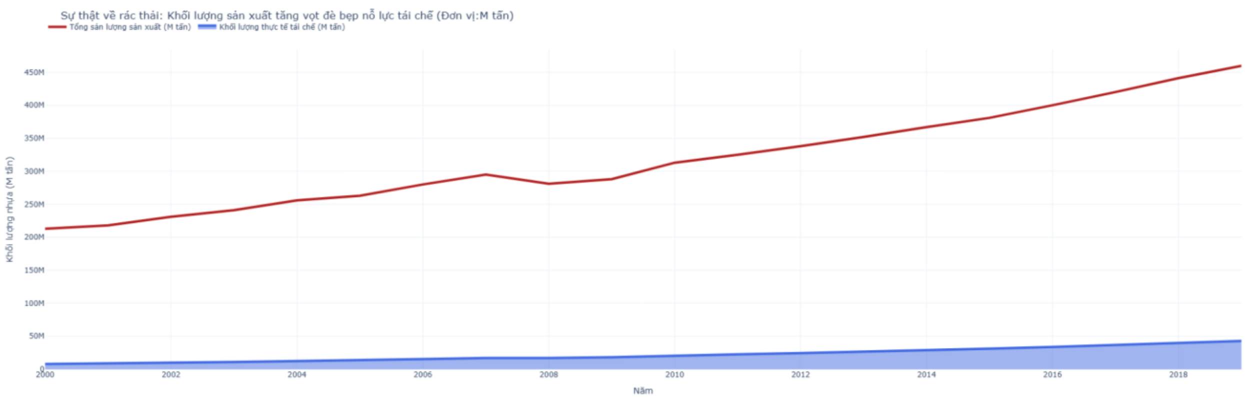
This stage represents our most significant technical improvement, transitioning from raw data processing to Data Engineering:

- Entity Filtering: Utilizing the Pandas library to scan for ISO codes. We identified that the raw data contained aggregate groups (e.g., "Asia", "World"). We developed a filtering function to separate two distinct datasets: one consisting strictly of countries (for comparative analysis) and another for regional aggregates (for global trend analysis).

- Data Merging: Executed pd.merge across four CSV files based on a composite key (Code, Year). This resulted in a Master DataFrame allowing for correlation analysis between "Economic Capacity" and "Waste Processing Efficiency."
- Unit Standardization: Converted percentage columns (%) into absolute mass (Tons) by multiplying them by the total annual production. This accurately reflects the severity of the issue, which is often masked by percentage figures.

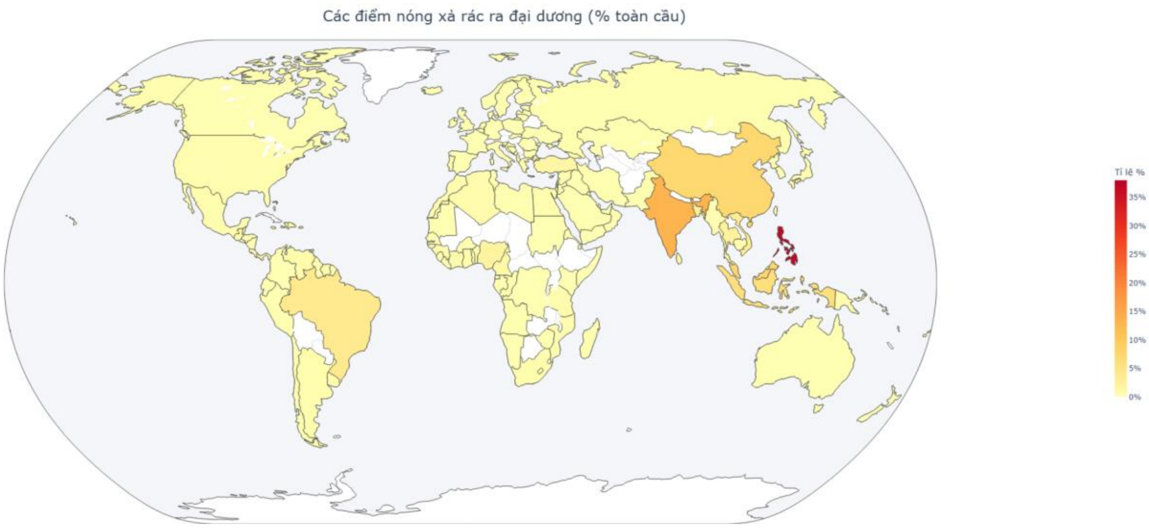
2.3. Visualization & in-depth analysis

Chart 1: Global Plastic Production Trends (1950 - 2019) – Uncontrolled Explosion



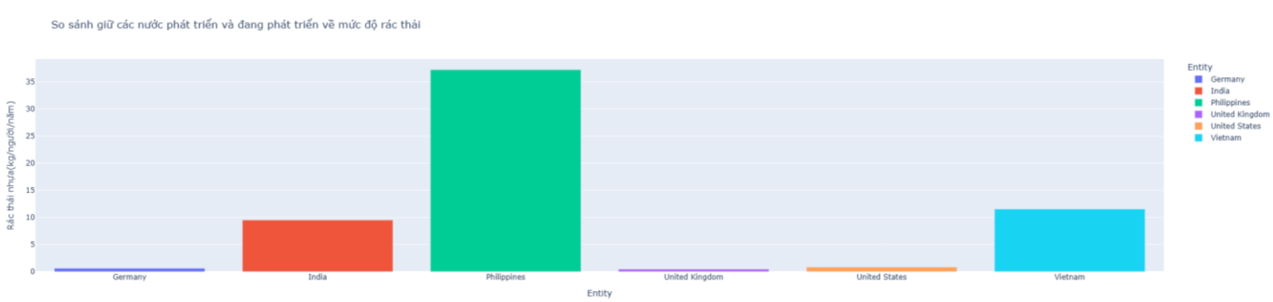
- Visualization: Line Chart with Area Fill to emphasize cumulative mass.
- In-depth Analysis: The graph identifies three phases:
    1. *1950 - 1970 (Emergence):* Plastic begins replacing wood and metal. Growth is slow but steady.
    2. *1970 - 2000 (Ubiquity):* The rise of single-use plastics. The gradient increases significantly.
    3. *2000 - 2019 (Explosion):* Production surges from 200 million to 460 million tons in just 19 years.
- Core Insight: This curve is exponential. Current waste processing infrastructure, built on the logic of previous decades, is "suffocating" under the weight of the new millennium's plastic volume. Plastic production has become a machine with no stop button.

Chart 2: Oceanic Leakage "Hotspots" – Who is Responsible?



- Visualization: Interactive Choropleth Map (Plotly).
- In-depth Analysis: The map reveals extreme polarization. While North America and Europe appear nearly "white" (negligible leakage), Southeast and South Asia show deep red hues (notably the Philippines, India, and Vietnam).
- Core Insight: There is a shift in the pollution burden. Developed nations "export" waste to developing nations with dense river systems but fragile collection infrastructure. Rivers act as "conveyor belts" carrying waste directly to the ocean. Conclusion: To save the oceans, we must intercept waste at the river mouths in Asia.

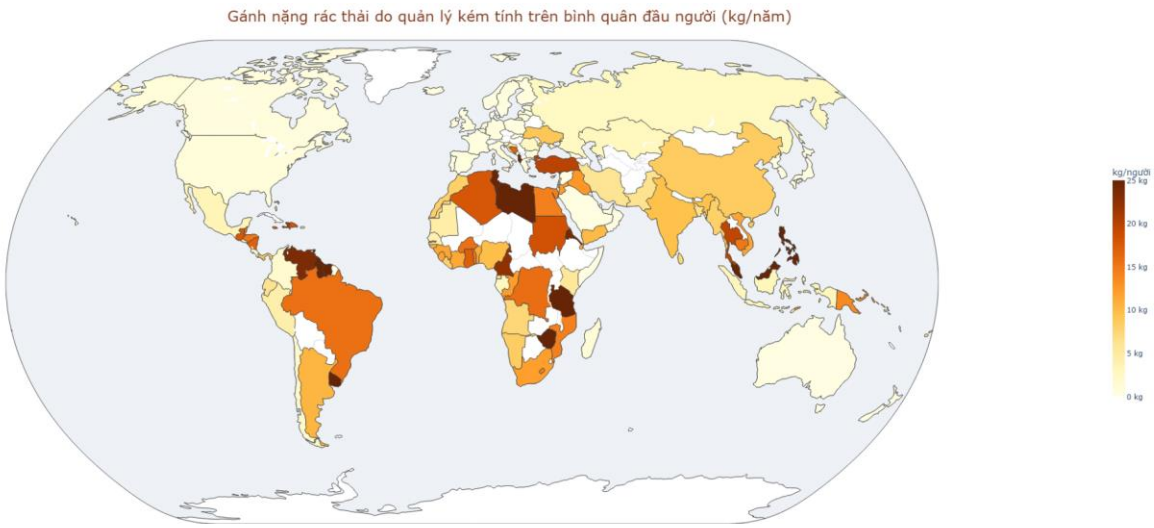
Chart 3: The Truth about Recycling – Solution or Smoke Screen?



- Visualization: Stacked Bar Chart depicting the "Fate of Plastic."
- In-depth Analysis: The gaps within the segments reveal a heartbreaking reality:
    1. Recycling rates account for only 9-15% and have grown extremely slowly over decades.
    2. Mismanaged waste and Landfilling consistently dominate the share (over 70%).
- Core Insight: We are living in a Linear Economy disguised as "recycling." Plastic is created, used once, and discarded. Over-focusing on recycling promotion may be

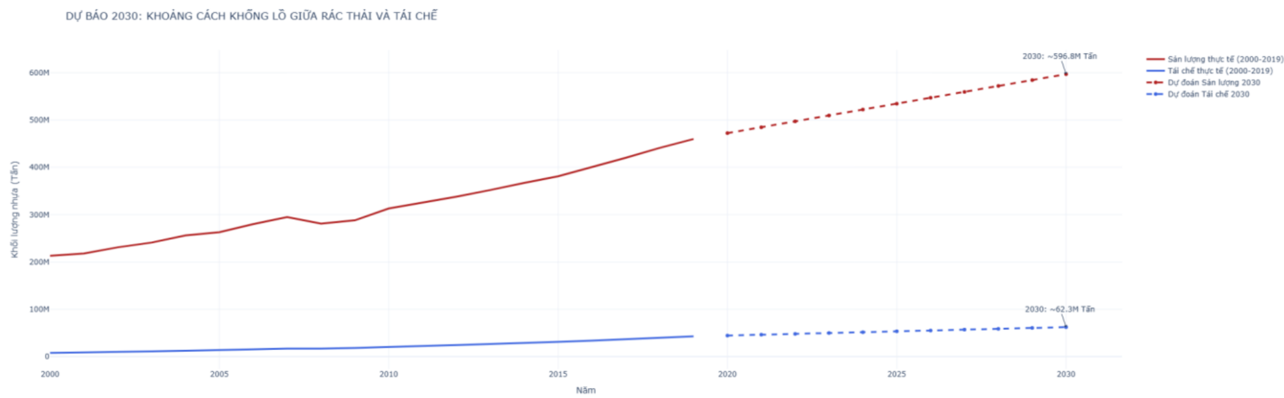
distracting from the real issue: we must Reduce production at the source rather than just attempting to treat the symptoms.

Chart 4: The Infrastructure Paradox – Why are Rich Nations Clean?



- Visualization: Scatter Plot with a Trendline.
- In-depth Analysis: This chart highlights the correlation between wealth and waste processing efficiency:
    1. *Developed Nations (USA, Japan, EU):* Extremely high plastic consumption but mismanaged waste is nearly zero.
    2. *Developing Nations:* Lower consumption but extremely high per capita leakage into the environment.
- Core Insight: Plastic pollution is a direct result of "Infrastructure Lag." As a nation transitions from low to middle income, plastic consumption grows faster than the capacity to build processing plants. Plastic only becomes "pollution" when it escapes a closed management system.

Chart 5: 2030 Forecast – Toward the "Point of No Return"



- Visualization: Prediction/Forecasting Plot using Regression techniques.
- In-depth Analysis: Based on an average growth rate of 4.1%/year, the model shows that by 2030, global production will reach nearly 600 million tons/year.
- Core Insight: Without a revolutionary shift in materials (bioplastics) or an international treaty mandating a reduction in virgin plastic, leakage volumes by 2030 will exceed the natural recovery capacity of oceanic ecosystems. This is a "Red Alert" for the future.

3. COMPARISON & IMPROVEMENT

During this semester, our two members (Hao Khang and Quoc Huy) completed two Midterm projects in entirely different fields (Traffic and Hydrology). Therefore, this Final Project on "Plastic Pollution" is not only a new topic but a culmination and upgrade of the best skills from both previous projects.

3.1. Comparison with Nguyen Hao Khang's Midterm Traffic Accidents

- **Problem Definition:**
  - *Midterm:* Limited to "What? and Where?" (e.g., Which state has the most accidents?).
  - *Final:* Shifted to critical "Why?" thinking—decoding the "Infrastructure Paradox" of why rich nations consume more yet poor nations suffer more pollution.
- **Data Engineering:**
  - *Midterm:* Used a single flat dataset.

    - *Final:* Upgraded to **Data Merging**. Handled conflicts across 4 CSV files, standardized ISO Codes, and performed time-series alignment—a skill not present in the Midterm.
- **Depth of Insights:**
    - *Midterm:* Observational conclusions (Bad weather causes accidents).
    - *Final:* Strategic conclusions (Intercepting waste at Asian river mouths).

3.2. Comparison with Nguyen Cao Quoc Huy's Midterm (Sea Level Fluctuations)

- **Data Scale & Dimensionality:**
    - *Midterm:* Narrow data focused on a specific area (San Francisco) with physical variables (wind, current).
    - *Final:* Expanded to a **Global Scale**. Managing data for over 190 countries required much more complex filtering and normalization than monitoring a single station.
- **Visualization Complexity:**
    - *Midterm:* Used basic static charts (Line, Scatter).
    - *Final:* Implemented an **Interactive Choropleth Map**. This allows users to interact directly with spatial data—a significantly more advanced visualization skill.
- **Storytelling:**
    - *Midterm:* Functioned as a technical weather forecast bulletin.
    - *Final:* Functioned as a **Policy Analysis Report**, combining historical forecasting with management evidence to provide actionable environmental recommendations.

4. DETAILED CONCLUSION & INSIGHTS

4.1. The Failure of Waste Management under Production Pressure

Results from Questions 1 and 3 indicate a staggering reality: while production grows exponentially, recycling remains stagnant.

- **Conclusion:** We are facing an **"Infrastructure Exhaustion"** state. Traditional methods like landfilling and incineration can no longer handle the load, leading to massive volumes of "Mismanaged Waste."

4.2. Decoding the Economic-Infrastructure Paradox

By cross-referencing country groups (Question 4):

- **Insight:** Plastic pollution is not an "awareness" issue but an **"Infrastructure Capacity Gap"** issue. This debunks the idea that individual awareness is enough; instead, investing in closed-loop collection and processing systems is the master key.

4.3. Strategic "Hotspots" on the Ocean Map

Interactive data (Question 2) shows over 80% of ocean plastic leaks from Asian rivers.

- **Conclusion:** Marine pollution is highly geographically concentrated. International efforts should pivot from costly ocean cleanups to intercepting waste at major river mouths in Southeast and South Asia.

4.4. 2030 Vision and the "Red Alert"

Based on the 2030 forecast (Question 5):

- The world faces a **"Point of No Return"** where plastic volume exceeds ecosystem recovery.
- Relying on recycling is an "illusion" if virgin plastic production remains unchecked.

4.5. Evidence-Based Recommendations

1. **Source Management:** Implement excise taxes or production quotas on virgin plastic to force a shift toward alternative materials.
2. **Transnational Infrastructure Support:** Developed nations should transfer technology and funding to build waste systems in identified "hotspots."
3. **Life Cycle Optimization:** Transition from a "Take-Make-Waste" model to a true **Circular Economy** where plastic is designed for multi-use from the outset.

Closing Remarks: This final project successfully transformed raw, disjointed data into a profound and persuasive narrative. The significant improvement over the Midterm lies not just in Python coding or interactive design, but in the **analytical mindset** used to challenge and solve an urgent, real-world human crisis.